

Building an App Using Android Jetpack Compose

Android Jetpack Compose is a new way of creating an Android layout; it eases the difficulties faced in traditional layout creation and maintenance. This article is a simple demonstration of how to use Jetpack Compose to build a sample app along with code based on Kotlin.



Jetpack Compose is a UI kit made for declarative programming in Android. With Jetpack Compose, you can avoid complicated xml layout files, but programmatically design the screens. The UI is built using composable functions in Jetpack Compose. In this tutorial, we will use it to make a simple application that will help users make a bio for themselves.

Figure 1 shows what our app looks like when it is opened. It should show an *Edit* button at the top. The profile picture should be displayed below the *Edit* button. Below this picture, the name, about, email, and the phone number are displayed. The app provides options to edit all the details by clicking the *Edit* button.

In editing mode (Figure 2), the app will display an *Edit* button on the profile picture; all other fields become editable, and a *Save* button is displayed.

On clicking the *Edit* button on the profile picture, the app will display an option to pick a profile picture from the list, as shown in Figure 3.

Creating the app

Open the Android Studio (Android Studio Dolphin or later). Go to it, then click *File -> New -> New Project*

-> *Empty Compose Activity*. Give application name, package name, save location and minimum SDK. API 21 is given as the minimum SDK.

String resources

Create a *string.xml* file with all of the required strings in our code. As per the UI, we require the following

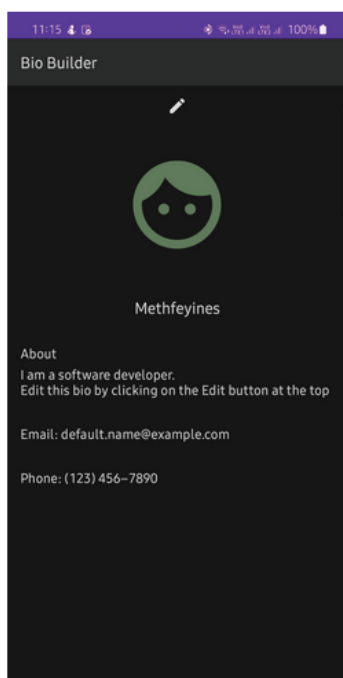


Figure 1: The Bio app

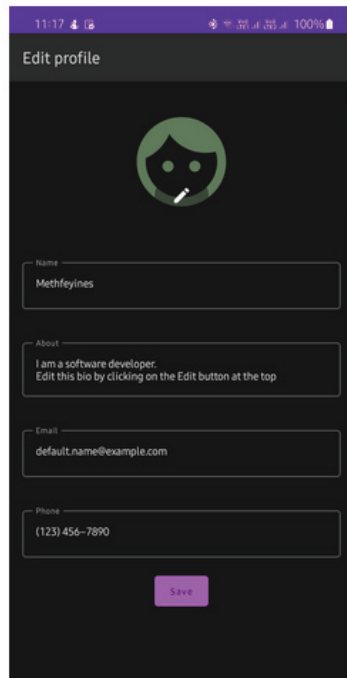


Figure 2: The app in editing mode

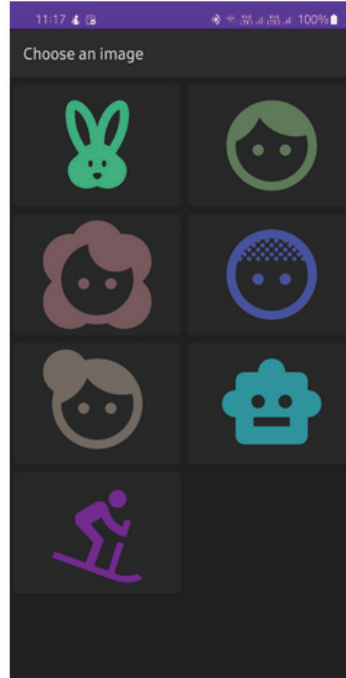


Figure 3: Selecting the profile picture